ATM Machine Simulation By Using C++

This presentation covers the development of a simple ATM machine simulation program using C++. It demonstrates essential programming concepts such as functions, pointers, loops, and user interaction through menus. The goal is to build an understanding of how C++ can be applied to create practical, interactive software systems that mimic real-world operations.

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Program Objectives

Display Balance

Show the user's current account balance clearly and accurately with every transaction.

Withdraw Money

Enable users to withdraw funds with validation to prevent overdrafts.

Input Validation

Check for sufficient balance before processing withdrawals to ensure data integrity.

Menu and Loop

Use menus and loops to allow multiple transactions until the user decides to exit.

Tools and Concepts Used

Functions

Organize code into reusable blocks that handle different operations like displaying balance or withdrawing cash.

Pointers

Modify the account balance directly from within functions by passing pointers, reflecting real-time updates.

While Loop

Keep the program running by repeatedly presenting the menu until the user opts to quit.

Switch Statement

Efficiently manage user inputs by directing program flow based on menu selections.

Program Flow and User Interaction

1. **Display Menu**

   Show options like check balance, withdraw, and exit to the user.

2. **Get User Input**

   Accept and validate the user's choice to ensure proper handling.

3. **Process Request**

   Using switch statements, perform the chosen action and update the balance if needed.

4. **Repeat or Exit**

   Loop the menu until the user chooses to end the session.

Key Takeaways and Next Steps

Practical Application

This project reinforces core C++ concepts in a meaningful context, improving programming skills.

Expand Functionality

Consider adding features like deposits, PIN security, or transaction history for greater realism.

Learning Path

Build on this simulation by exploring object-oriented programming, file handling, and error management.